TEXORA · LECTURE SHEET

Others Special Fibres

Fibres engineered for extremes: aramids for heat and ballistic protection, carbon fibre for stiffness, glass fibre for fireproof reinforcement, gel-spun polyethylene for strength at feather weight, and the smaller specialities around them.

TE-2108 · MAN-MADE TEXTILE FIBERS · TOPIC 4 · LEVEL 2 TERM 1

1. What "special" means

The commodity fibres of Topic 3 compete on price and general usefulness. **Special (high-performance) fibres** are bought for one extreme property — strength-to-weight, heat resistance, chemical inertness — at prices ten to a hundred times higher, in small tonnages for technical textiles: protective clothing, aerospace and automotive composites, ropes and cables, filtration, ballistics. They share a structural secret: extremely rigid and/or highly oriented chains, often aromatic rings in the backbone, that ordinary fibres cannot match.

2. Aramids — the aromatic polyamides

Aramids are polyamides in which at least 85 % of the amide links join **aromatic rings** directly. The ring-stiffened chains cannot fold, so the fibres have exceptional thermal stability and strength — and cannot be melt-processed; they are solution-spun from concentrated sulfuric acid or polar solvents.

2.1 Meta-aramid (Nomex type)

Poly(*m*-phenylene isophthalamide): the meta links put kinks in the chain, limiting strength to a useful ≈ 4–6 g/den but giving superb **heat and flame behaviour**: it does not melt, chars above roughly 370 °C, and keeps working in flame and radiant heat. Firefighters' turnout gear, racing drivers' suits, hot-gas filter bags, electrical insulation paper.

2.2 Para-aramid (Kevlar, Twaron types)

Poly(*p*-phenylene terephthalamide): the para links keep the chain ruler-straight; spun from a liquid-crystalline (lyotropic) sulfuric-acid dope, the chains emerge almost perfectly aligned and hydrogen-bonded into sheets. Tenacity ≈ 20 g/den and above (tensile strength around 3 GPa) with a modulus rivalling steel at one-fifth of steel's density — the ballistic and reinforcement fibre: bullet-resistant vests, cut-resistant gloves, tyre and hose reinforcement, fibre-optic cable strength members, aerospace

composites. Weaknesses: poor UV resistance (it self-screens and yellows, losing strength), poor compressive strength, moisture regain ≈ 4–7 % that slightly plasticises it.

> **WORKED EXAMPLE — WHY "STRENGTH-TO-WEIGHT" FAVOURS PARA-ARAMID**
>
> Compare specific strength (tensile strength ÷ density):
>
> Para-aramid: 3.0 GPa ÷ 1.44 g/cm³ = **2.08** (GPa per g/cm³).
>
> High-tensile steel wire: 2.2 GPa ÷ 7.85 g/cm³ = **0.28**.
>
> Ratio = 2.08 ÷ 0.28 ≈ **7.4** — weight for weight, the aramid is roughly seven times as strong as steel wire. That is the whole commercial story of the fibre: a vest or cable can carry steel-like loads at a fraction of the mass. (Absolute strength is steel's; only the specific value is the aramid's.)

3. Carbon fibre

Production: about 90 % of carbon fibre starts as **PAN precursor** filament (Topic 3's acrylic, specially pure). Three furnace stages convert it: (i) **stabilisation** in air at 200–300 °C — the PAN chains cyclise into a ladder structure that will not melt; (ii) **carbonisation** in inert gas at 1000–1500 °C — nitrogen, hydrogen and other elements are driven off, leaving ≈ 90 %+ carbon as fused ribbon-like graphitic layers; (iii) optional **graphitisation** at 2000–3000 °C for maximum modulus. Surface treatment and sizing follow so the fibre bonds to resin.

Properties: tensile strength roughly 3.5–7 GPa and modulus 230–600 GPa depending on grade, at density ≈ 1.75–1.95 g/cm³; elongation only 1–2 % (brittle — it is a ceramic-like filament); superb fatigue and creep resistance; electrically conductive; does not melt or burn in normal use. Aircraft structures, racing cars and bicycles, pressure vessels, wind-turbine spars, sports goods — and technical fabrics where conductivity or reinforcement is wanted. Carbon fibre fabrics are woven gently: the filaments hate sharp bending and abrasion.

4. Glass fibre

Production: silica sand, limestone and other minerals are melted at about 1400 °C (E-glass composition) and drawn through platinum bushings with hundreds of holes at high speed, then sized and gathered. The fibre is simply frozen glass in filament form.

Properties: density ≈ 2.5 g/cm³, tenacity ≈ 8–15 g/den as pristine filament (strength falls with surface damage and moisture), does not burn, soften until ≈ 800 °C+, immune to rot and most chemicals, cheap. Brittle in flexing and irritating to skin, so its textile role is technical: reinforcement in composites (boats, boards, pipes), insulation, fire blankets, filtration, and the carrier fabrics of coated membranes.

5. UHMWPE — gel-spun polyethylene

Ultra-high-molecular-weight polyethylene (chains of several million g/mol) cannot be melt-spun — the melt barely flows. In **gel spinning** it is dissolved in a hot solvent to a semi-dilute solution, extruded, cooled to a gel filament, and drawn to extreme ratios (30–100×) after solvent extraction. The chains unfold and align almost perfectly, giving tenacity around 30–40 g/den (strength ≈ 3–4 GPa) at density 0.97 g/cm³ — the strongest fibre per weight in commerce, and it floats. Sold as Dyneema/Spectra types for cut-resistant gloves, ballistic panels, mooring ropes and medical sutures. Limits: melts ≈ 145–150 °C, creeps under sustained load, and nothing sticks to it (a composite-bonding headache).

6. The supporting cast

FIBRE	EXTREME PROPERTY	TYPICAL USES
PTFE fibre	Total chemical inertness, −200 to +260 °C service, near-zero friction	Chemical filtration, pump packing, high-temperature sewing thread, membrane fabrics (expanded PTFE)
PPS (polyphenylene sulfide)	Melts ≈ 285 °C, inherent flame retardancy, chemical resistance	Hot-gas filtration in power and cement plants
Melamine fibre (Basofil type)	Does not melt, low thermal conductivity, flame barrier	Fire-blocking layers in seats and bedding, heat-protective clothing blends
PBI	Highest heat/flame stability of organic textile fibres; does not burn in air	Fire-service and space applications — at the highest price in this table
Conductive fibres (metal, carbon-loaded)	Electrical conductivity built into a textile yarn	Antistatic workwear, EMI shielding, heating textiles, sensors (e-textiles, EEE-2203 Topic 7)
Optical fibre (silica or PMMA)	Guides light along the filament	Data cables; decorative and sensing textiles
Ceramic fibres (alumina, silicon carbide)	Service above 1000 °C	Furnace insulation, high-temperature seals and composites

FIBRE	TENSILE STRENGTH (APPROX.)	DENSITY (G/CM³)	HEAT LIMIT	HEADLINE WEAKNESS

Para-aramid	≈ 3.0 GPa	1.44	Chars ≈ 425 °C+, no melt	UV degradation
Meta-aramid	≈ 0.6–0.9 GPa	1.38	Chars ≈ 370 °C+, no melt	Moderate strength only
Carbon	3.5–7 GPa	1.75–1.95	Oxidises in air above ≈ 400–500 °C	Brittle, 1–2 % elongation
E-glass	≈ 2–3.5 GPa (pristine)	≈ 2.5	Softens ≈ 800 °C+	Flex fatigue, skin irritation
UHMWPE	≈ 3–4 GPa	0.97	Melts ≈ 150 °C	Creep; nothing bonds to it

Glossary

Aramid — aromatic polyamide; meta and para forms differ by ring linkage geometry. **Lyotropic** — liquid-crystalline in solution (para-aramid dope aligns before it is even drawn). **Precursor** — the starting filament (usually PAN) converted into carbon fibre. **Carbonisation** — heating in inert gas to drive off non-carbon elements. **Gel spinning** — spinning from a solution that gels on cooling, allowing extreme draw of UHMWPE. **Specific strength** — strength divided by density; the correct figure of merit for weight-critical uses.

Key points

• Special fibres trade price for one extreme property; rigid aromatic or perfectly aligned chains are the common structural recipe.
• Meta-aramid sells heat protection; para-aramid sells specific strength (≈ 7× steel wire per weight) and fears UV.
• Carbon fibre = PAN, stabilised, carbonised (optionally graphitised): highest stiffness, brittle, conductive.
• Glass is mineral, cheap, fireproof reinforcement; UHMWPE is the strongest fibre per weight but melts low and creeps.
• No special fibre is general-purpose: each carries a headline weakness the designer must engineer around.

Practice

1. A firefighter's suit must not melt, must char-protect, and is laundered industrially. Choose between meta-aramid, para-aramid and UHMWPE, and justify the rejected two.

2. Calculate the specific strength of E-glass at 2.5 GPa and 2.5 g/cm³, and compare it with para-aramid's 2.08 from the worked example. Why is glass still the volume leader in composites despite the answer?
3. Put the carbon-fibre process stages in order and state the chemical job of each: carbonisation, stabilisation, sizing/surface treatment, graphitisation (optional).
4. Why is para-aramid spun from sulfuric acid solution instead of from a melt?
5. A mooring rope is specified in UHMWPE rather than nylon. Give two advantages for this use and the one service limit the harbour engineer must respect.

Answer key

1. **Meta-aramid.** Para-aramid is stronger but its talent is wasted here and it costs more; UHMWPE is disqualified absolutely — it melts at ≈ 150 °C, the exact failure the suit exists to prevent. Meta-aramid does not melt, chars protectively, and survives repeated hot laundering.
2. Specific strength = 2.5 ÷ 2.5 = **1.0** — about half of para-aramid's 2.08. Glass still leads in volume because composites for boats, boards and pipes are stiffness- and price-driven, not weight-critical: glass costs a small fraction of aramid or carbon per kilogram and is far easier to process.
3. Stabilisation (air, 200–300 °C): cyclise PAN into a heat-proof ladder so it survives later stages. Carbonisation (inert, 1000–1500 °C): expel non-carbon elements, form the graphitic ribbons — the strength/stiffness appear. Graphitisation (2000–3000 °C, optional): perfect the graphitic order for maximum modulus. Surface treatment + sizing: oxidise the surface lightly and coat it so resin wets and bonds to the fibre.
4. Its rigid, hydrogen-bonded aromatic chains decompose below any temperature at which they could flow — there is no melt. The acid dissolves the chains (breaking the hydrogen bonds) into a liquid-crystalline dope whose inherent alignment is then preserved through the spinneret.
5. Advantages: highest strength per weight (smaller, lighter rope for the same load, and it floats if dropped overboard); near-zero water absorption and excellent chemical/abrasion behaviour in seawater. Limit: creep and the ≈ 150 °C melting point — sustained high loads permanently elongate it, and it must be kept away from hot surfaces and friction heating.
